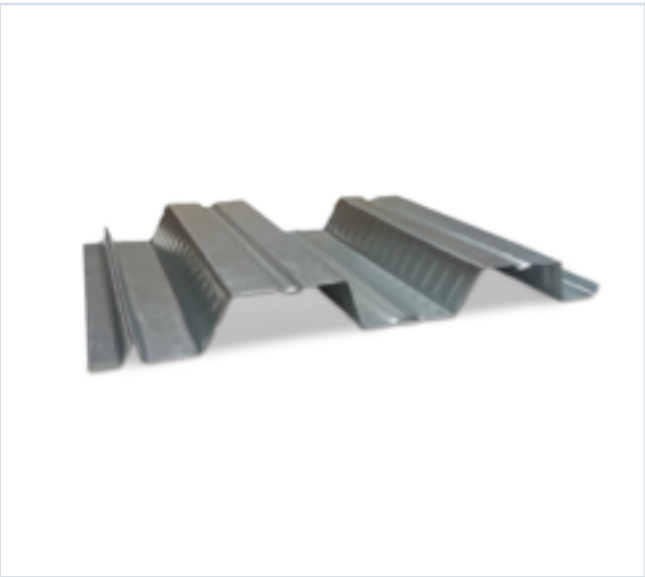


DECKING PRODUCT SUBMITTAL

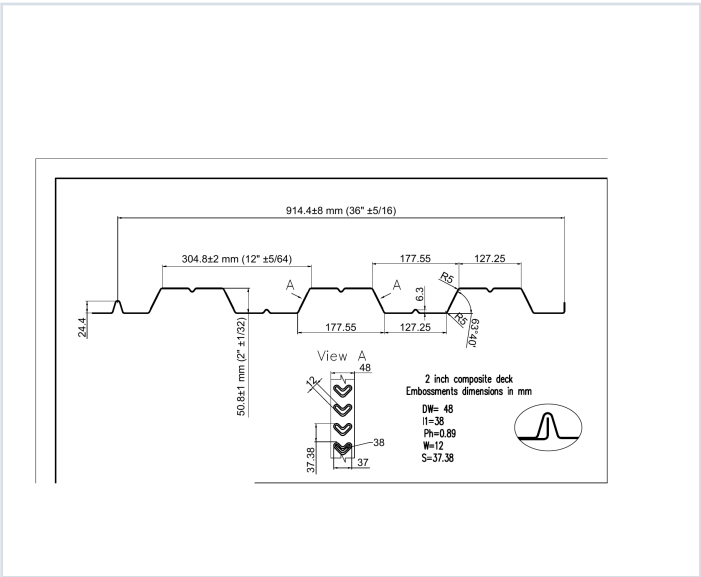
2CD-FY50-20-G90

PRODUCT	GRADE	GAUGE	COATING
2" Composite Deck	Grade 50	20 ga	G90

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 20 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0358	1.9	50	0.329	0.336	9860	10060	0.393	0.373

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 54, 55, 56, 57, 58, 59, 60, 61
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

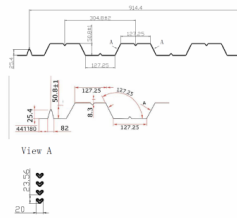
2CD-FY50-20-G90 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 54

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (pcf)	E _s (ksi)	S _x (in ³)	S _y (in ³)	ASD (E _s + 16E _t)	M _u (ft-k)	M _y (ft-k)	I _x (in ⁴)	I _y (in ⁴)
22	0.0205	16	50	0.247	0.254	2427	2000	2000	0.31	0.290
20	0.0358	19	50	0.328	0.336	3860	3000	3000	0.393	0.373
18	0.0474	25	50	0.485	0.500	5470	4500	4500	0.548	0.530
16	0.0598	32	50	0.645	0.664	7420	6000	6000	0.703	0.683

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10C and AISC 360-10D.

Shear and Web Crippling

Gage	V _u (kips)	Web Crippling (R _u /V _u)		Web Crippling (R _u /V _u)	
		End Bearing	Interior Bearing	End Bearing	Interior Bearing
22	388	434	476	529	627
20	278	295	325	364	427
18	205	218	242	278	325
16	160	170	190	218	258

Note: All section properties and ASD flexural strengths are calculated in accordance with ANSI/AISC 360-10, AISC 360-10C and AISC 360-10D.

STEEL CATALOG PAGE 55

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	137	133	127	121	115	109	103	97	91	85	79
	20	183	178	171	164	157	150	143	136	129	122	115
	18	273	267	258	249	240	231	222	213	204	195	186
	16	358	351	340	329	318	307	296	285	274	263	252
Double	22	131	127	121	115	109	103	97	91	85	79	73
	20	186	181	173	165	157	149	141	133	125	117	109
	18	277	271	261	251	241	231	221	211	201	191	181
	16	367	360	348	336	324	312	300	288	276	264	252
Triple	22	176	171	163	155	147	139	131	123	115	107	99
	20	243	237	228	219	210	201	192	183	174	165	156
	18	346	339	328	317	306	295	284	273	262	251	240
	16	446	438	426	414	402	390	378	366	354	342	330

Note: 1. All section properties and ASD (E_s + 16E_t) uniform loads are calculated in accordance with ANSI/AISC 360-10, AISC 360-10C and AISC 360-10D.
2. Loads shown in tables are uniformly distributed superimposed loads in psf. Span lengths include center-to-center spacing of supports. Tabulated loads shall not be increased by assuming clear span dimensions.
3. Bending Moment formula used for flexural stress limitations are: Single and Two Span: $M_u = \frac{wL^2}{8}$; Three Span: $M_u = \frac{wL^2}{10}$.

Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

Uniform Superimposed Service Load that Causes L/240 Deflection (PSF)

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	13	13	12	11	10	9	8	7	6	5	4
	20	18	17	16	15	14	13	12	11	10	9	8
	18	27	26	25	24	23	22	21	20	19	18	17
	16	36	35	34	33	32	31	30	29	28	27	26
Double	22	13	12	11	10	9	8	7	6	5	4	3
	20	18	17	16	15	14	13	12	11	10	9	8
	18	27	26	25	24	23	22	21	20	19	18	17
	16	36	35	34	33	32	31	30	29	28	27	26
Triple	22	17	16	15	14	13	12	11	10	9	8	7
	20	24	23	22	21	20	19	18	17	16	15	14
	18	34	33	32	31	30	29	28	27	26	25	24
	16	44	43	42	41	40	39	38	37	36	35	34

Note: For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.667.

PUBLISHED PERFORMANCE DATA

2CD-FY50-20-G90 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

STEEL CATALOG PAGE 56

Normal Weight Concrete (145 pcf)				Lightweight Concrete (115 pcf)			
Maximum Unshored Clear Span				Maximum Unshored Clear Span			
400 (#200) 39 PSF	2x12x22 ga	7'-8"	9'-1"	9'-3"	2x12x22 ga	8'-6"	9'-9"
	2x12x20 ga	9'-3"	10'-6"	10'-10"	2x12x20 ga	10'-0"	11'-3"
	2x12x18 ga	10'-5"	12'-9"	12'-5"	2x12x18 ga	11'-2"	13'-9"
	2x12x16 ga	11'-4"	14'-2"	13'-4"	2x12x16 ga	12'-0"	14'-8"
450 (#250) 45 PSF	2x12x22 ga	7'-7"	8'-8"	8'-10"	2x12x22 ga	8'-3"	9'-5"
	2x12x20 ga	8'-10"	10'-0"	10'-4"	2x12x20 ga	9'-7"	10'-10"
	2x12x18 ga	9'-11"	12'-2"	12'-0"	2x12x18 ga	10'-9"	13'-3"
	2x12x16 ga	10'-11"	13'-8"	12'-10"	2x12x16 ga	11'-8"	14'-6"
500 (#300) 51 PSF	2x12x22 ga	7'-3"	8'-4"	8'-6"	2x12x22 ga	7'-8"	9'-1"
	2x12x20 ga	8'-6"	9'-7"	9'-11"	2x12x20 ga	9'-3"	10'-6"
	2x12x18 ga	9'-7"	11'-8"	11'-8"	2x12x18 ga	10'-5"	12'-9"
	2x12x16 ga	10'-6"	13'-3"	12'-6"	2x12x16 ga	11'-4"	14'-2"
550 (#350) 57 PSF	2x12x22 ga	6'-11"	8'-0"	8'-2"	2x12x22 ga	7'-7"	8'-9"
	2x12x20 ga	8'-3"	9'-3"	9'-6"	2x12x20 ga	8'-11"	10'-1"
	2x12x18 ga	9'-3"	11'-3"	11'-4"	2x12x18 ga	10'-0"	12'-4"
	2x12x16 ga	10'-1"	12'-9"	12'-2"	2x12x16 ga	11'-0"	13'-11"
600 (#400) 63 PSF	2x12x22 ga	6'-9"	7'-9"	7'-10"	2x12x22 ga	7'-5"	8'-6"
	2x12x20 ga	7'-8"	8'-11"	9'-2"	2x12x20 ga	8'-8"	9'-9"
	2x12x18 ga	8'-11"	10'-10"	11'-0"	2x12x18 ga	9'-9"	11'-10"
	2x12x16 ga	9'-9"	12'-4"	11'-10"	2x12x16 ga	10'-8"	13'-6"
650 (#450) 69 PSF	2x12x22 ga	6'-8"	7'-6"	7'-7"	2x12x22 ga	7'-2"	8'-3"
	2x12x20 ga	7'-9"	8'-7"	8'-10"	2x12x20 ga	8'-5"	9'-5"
	2x12x18 ga	8'-8"	10'-6"	10'-9"	2x12x18 ga	9'-5"	11'-6"
	2x12x16 ga	9'-6"	11'-8"	11'-7"	2x12x16 ga	10'-4"	12'-4"

Note
With crippling and clear span not shown accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 61

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	375	330	290	260	230
4.5	35	400	400	400	400	395	397	384
5	39	400	400	400	400	400	370	337
5.5	44	400	400	400	400	400	400	393
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	290	280	272	257	243	230	220	210
4.5	296	298	290	280	276	260	247	236
5	304	295	290	277	268	259	249	240
5.5	304	300	291	285	272	262	254	246
6	307	307	304	304	298	290	284	276
6.5	307	307	307	304	304	298	290	284

16 ga Lightweight Concrete (115 pcf, F_c = 3,000 psi)

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	400	400	392	323	290
4.5	35	400	400	400	400	400	390	360
5	39	400	400	400	400	400	400	400
5.5	44	400	400	400	400	400	400	400
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	260	256	250	236	229	214	201	190
4.5	267	267	261	246	238	228	214	203
5	277	277	271	254	246	236	221	210
5.5	284	284	278	260	250	239	224	213
6	290	290	284	264	254	243	228	217
6.5	290	290	284	264	254	243	228	217